

## Total Energies of O:

|                  | aug-cc-pCV2Z   | aug-cc-pCV3Z   | aug-cc-pCV4Z   | aug-cc-pCV5Z   | aug-cc-pCV6Z   | aug-cc-pCV7Z   | aug-cc-pCV8Z   |
|------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| UHF              | -74.796 718 39 | -74.813 095 62 | -74.817 689 84 | -74.818 844 59 | -74.818 949 07 | -74.818 977 27 | -74.818 979 42 |
| CCSD             | -74.963 321 71 | -75.031 929 31 | -75.051 321 99 | -75.057 488 48 | -75.059 306 76 | -75.060 624 40 | -75.061 137 72 |
| CCSD(T)          | -74.964 542 90 | -75.035 772 11 | -75.053 445 10 | -75.058 937 45 | -75.062 712 94 | -75.058 428 31 | -75.059 453 98 |
| CCSDT            | -74.966 288 38 | -75.036 120 56 | -75.056 006 46 | -75.062 252 43 | -75.064 106 43 | -75.065 384 95 | -75.065 908 30 |
| CCSDT(Q)/B       | -74.966 471 22 | -75.036 195 95 | -75.056 183 09 | -75.062 441 19 | -75.064 264 13 | -              | -              |
| CCSDTQ           | -74.966 441 22 | -75.036 199 10 | -75.056 173 82 | -75.062 424 00 | -              | -              | -              |
| CCSDTQ(P)/B      | -74.966 446 07 | -75.036 196 64 | -75.056 176 92 | -              | -              | -              | -              |
| CCSDTQP          | -74.966 445 87 | -75.036 198 65 | -              | -              | -              | -              | -              |
| CCSDTQP(H)/B     | -74.966 445 91 | -              | -              | -              | -              | -              | -              |
| CCSDTQPH         | -74.966 445 91 | -              | -              | -              | -              | -              | -              |
| CCSDTQPH(S)/B    | -74.966 445 91 | -              | -              | -              | -              | -              | -              |
| CCSDTQPHS        | -74.966 445 92 | -              | -              | -              | -              | -              | -              |
| CCSDTQPHS(O)/B   | -74.966 445 92 | -              | -              | -              | -              | -              | -              |
| FCI = CCSDTQPHSO | -74.966 445 92 | -              | -              | -              | -              | -              | -              |

Note: The results for aug-cc-pCV3Z were taken from CFOUR-MRCC with cctol=10